UNIVERSITY OF BAGHDAD

COLLEGE OF MEDICINE

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Organic nutrition; carbohydrate, protein, and fat

Department of family and community medicine

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Learning Objectives

- To recognize the dietary CHO classification and dietary fibers.
- To classify amino acid & protein , nitrogen balance and factors affecting it.
- To calculate recommended intakes of CHO, protein and fat.
- To list the importance of dietary fibers, CHO, protein, fat,& E.F.A.

Carbohydrate

Chemists describe the sugars as:

- **Monosaccharides (single sugars): Glucose, Fructose, Galactose**

Glucose: Most cells depend on glucose for their fuel to some extent:

- ✓ **The brain and CNS primarily but not exclusively** uses glucose as its energy source, but during prolonged fasting or starvation, it can adapt to use ketone bodies as an alternative fuel.
- ✓ **Red blood cells (RBCs) → exclusively depend on glucose** because they lack mitochondria and therefore cannot oxidize fatty acids or ketone bodies.

- **Disaccharides (double sugars): Sucrose, lactose, maltose.**
- **Polysaccharides:** compounds composed of chains of monosaccharide units(glucose). **3 types: glycogen, starch, and fibers.**

1. **Glycogen:** is a storage form of energy for **human** beings and **animals** **(The human body stores much of its glucose as glycogen in the liver& muscles).**
2. **Starch:** All starchy foods come from **plants** (Grains, legume and root vegetable) are the richest food source of starch.

Resistant starches: starches that escape the digestion in the small intestine and reach the colon, and partially fermented by the gut microflora.

Such as **oatmeal,** whole grains, beans, **legumes, unripe bananas**, cooked then chilled potatoes, pasta, and rice.

Health benefits:

1. Gut health and prebiotic effects.
2. Glycemic control → improve the insulin sensitivity,
3. Body weight management and body fat reduction → prolongs satiety
4. Prevention of cardiovascular disease → ↑ HDL, ↓ LDL and VLDL
5. Blood pressure control

3. Fibers: are found in all plant derived foods **(vegetables, fruits, whole grains, and legumes…).** Just as starch is, but in **fibers the sugar units are held together by bonds that human digestive enzymes cannot break.** Most dietary fibers providing **no or little energy (1.5 to 2.5 kcal/gm),** depending on the extent to which they digested by bacteria in the human intestine and can generate energy when metabolized.

Fibers divided into 2 general groups by their chemical & physical properties:

Soluble fibers	Insoluble fiber
Dissolve in water	Do not dissolve in water
Form gels (are viscous)	Do not form gels (are not viscous)
More readily digested by gut bacteria	Less easily digested by gut bacteria
Found in barley, legumes, fruits, oats, and vegetables,	Bran, strings of celery, the hulls of seeds, and the skins of corn kernels.
Associated with lower risks of chronic diseases.	Aid the digestive system by easing elimination.

- AI of fibers **14 grams per 1000 kcalorie** intake **(roughly 25-35gm/day)** and intake should consist of equal amount of soluble & insoluble fibers
- **Too much fiber is no better than too little** WHO recommends an **UL of 40 gm/day.**
- **Increases in dietary fiber intake should be made gradually.** A sudden boost in dietary fiber can lead to uncomfortable bloating, gas, and cramping and should be associated with an appropriate fluid intake of at least 8 glasses of water per day and physical activity.

Health benefits of fiber:

- ✓ Role in CVD → **lowering blood cholesterol** levels (soluble fiber).
- ✓ Promotes normal bowel function → **prevents constipation.**
- ✓ Increases satiety → **prevention of obesity.**
- ✓ Protects against disorders of the small and large intestines (e.g., **IBS, diverticulosis, and colon cancer).**
- ✓ **Slows glucose absorption** → reducing blood glucose spikes and insulin secretion.
- ✓ Stimulate **chewing → saliva flow and gastric juice.**

Prebiotic vs Probiotic

- **Prebiotic:** A non-digestible dietary carbohydrate that is selectively utilized by beneficial microorganisms in the colon, thereby promoting their growth and/or activity (such as Lactobacilli and Bifidobacteria), hence they acts fertilizers for these bacteria.
- **Dietary examples:whole wheat, onions, garlic, leeks, banana, and breast milk.**



Probiotics:

- **Probiotics:** are live microorganisms, when administered in adequate amounts, confer a health benefit on the host. Lacticobacili and bifidobacteria are the most common types of microbes used as probiotics.
- Probiotic enhance the balance between micrbiota community(eubiosis) and prevent or reverse intestinal dysbiosis.
- Dietary examples: **yogurt, fermented and unfermented milk, cottage cheese, sauerkraut and pickles, and breast milk.**

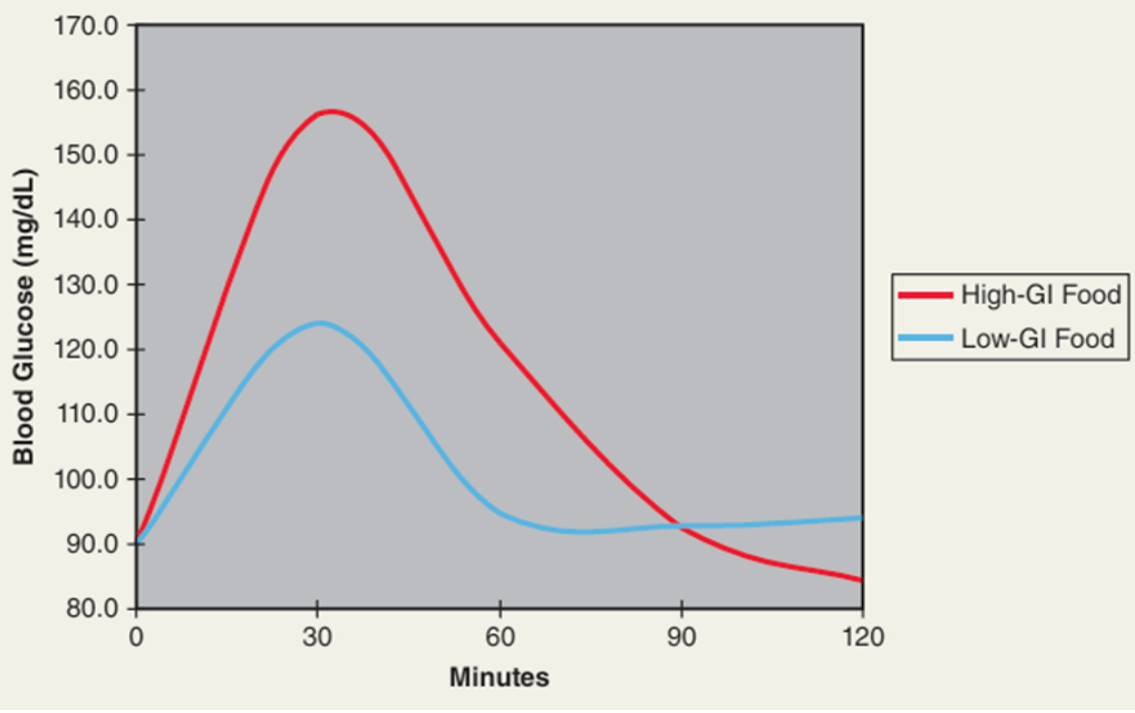
Dysbiosis and symbiosis ??

Glycemic Index (GI): is a scale that ranks carbohydrate containing foods by how much they raise blood glucose levels as compared with a specific amount (50 g) of a reference food (white bread or pure glucose).

The GI ranks carbohydrate containing foods from **0 to 100**

- ***Low GI Foods (55 or less) Choose Most Often, and preferable***
- ***Medium GI Foods (56-69) Choose Occasionally, and less preferable***
- ***High GI Foods (70+) Choose Less Often, and not preferable***

- Foods that produce **a higher peak** in blood sugar within 2 hours of eating them are given **a higher GI ranking**.
- Help in **controlling blood glucose levels**, specifically for individuals with diabetes.
- **GI= glycemic quality.**



Recommended dietary Allowance of CHO:

- The Acceptable Macronutrient Distribution Range (AMDR) of CHO **45-65% of total energy intake.**
- Refined sugar limited to **5-10%.**
- Minimum intake **130 gm/day** is needed to meet glucose needs and prevent ketosis.

The Dietary functions of CHO:

1. Major & cheapest source of energy
2. Protein sparing action.
3. Health benefit of fiber

Protein

Classification of a.a according to essentiality :

1. **Essential a.a.:** are a.a. that the human body **cannot synthesize** in sufficient amounts to meet its physiological needs and therefore must be obtained from the diet include **Histidine, Isoleucine, Leucine, Lysine, Methionine, Phenylalanine ,Threonine, Tryptophan, Valine**

2. **Non-essential a.a.:** are a.a. that the human body can synthesize in sufficient amounts to meet its physiological needs and therefore do not have to be obtained from the diet include **Alanine, Asparagine, Aspartic acid, Glutamic acid, Serine, Cysteine, Glutamine, Arginine, Tyrosine, Proline, Glycine**

3. Conditionally essential a.a.: are amino acids that the body can normally synthesize, but under certain physiological or pathological conditions (such as rapid growth, prematurity, severe illness, injury, or metabolic stress) the body may not produce sufficient amounts, so they must be obtained from the diet.

Cysteine, Glutamine, Arginine, Tyrosine, Proline, Glycine

Arginine is conditionally essential during periods of metabolic stress, such as malnutrition, sepsis, trauma, and recovery from surgery.

Classification of protein according to their quality:

1. Complete protein or High biological value protein:

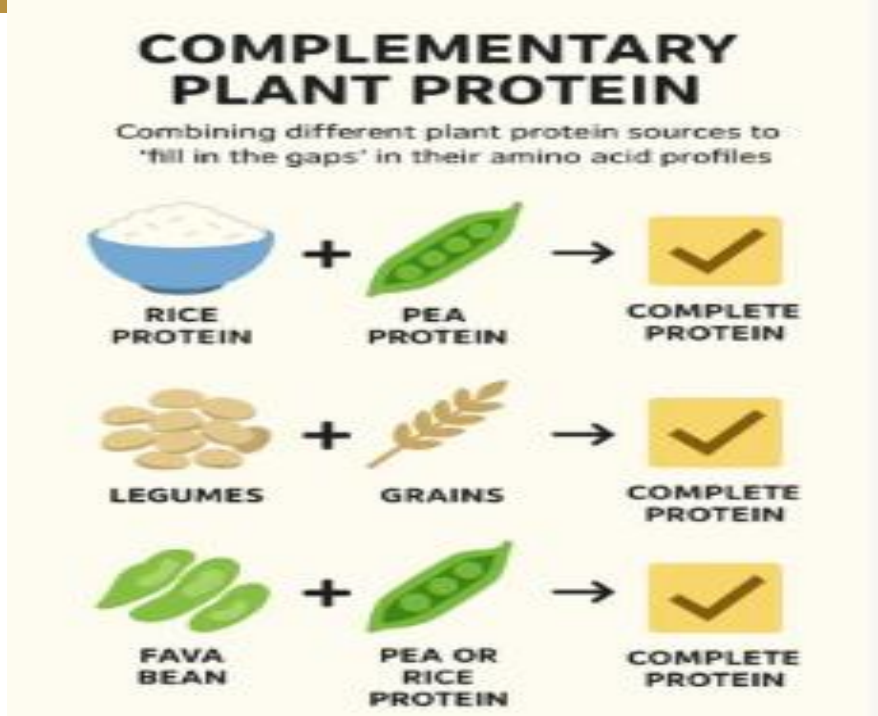
Contains all the 9 essential a.a. in sufficient amounts and appropriate proportions to support protein synthesis. **Most animal proteins are complete, except gelatin**

2. Incomplete protein: Lacks one or more of the 9 essential a.a. or contains them in insufficient amounts. It is mainly found in plant foods. **Soy protein is an exception because it is a high-quality, complete plant protein**





3. However, combining different plant protein sources, or combining plant proteins with a small amount of animal protein, can provide all the essential amino acids needed to support tissue growth and repair. This concept is known as **protein complementation**.



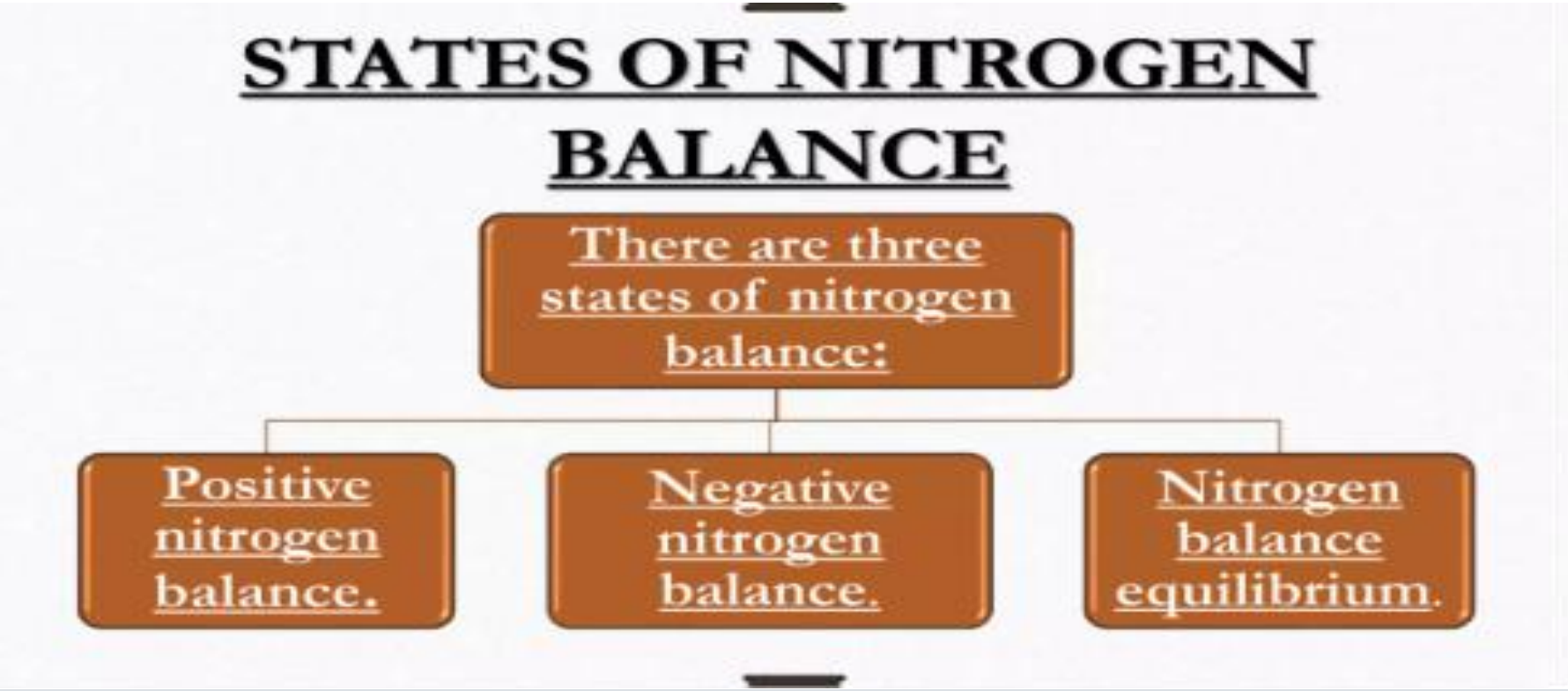
	Ile	Lys	Met	Trp
Legumes	✓	✓		
Grains			✓	✓
Together	✓	✓	✓	✓

Functions of proteins:

1. Proteins form structural parts of most body tissues
2. Proteins facilitate chemical reactions(enzymes).
3. Proteins transport substances such as lipids, vitamins, minerals, and oxygen around the body.
4. Proteins help to maintain the Fluid and electrolyte balance.
5. Proteins help maintain the acid–base balance (buffers).
6. Proteins defend the body against disease(antibodies).
7. Some, but not all protein, act as hormones.
8. Proteins provide some energy and glucose if needed.

What is the nitrogen balance?

- **Definition:** The difference between the amount of **nitrogen intake** and the amount of **nitrogen excreted or lost** from the body is called nitrogen balance.
- **Formula:** Nitrogen Balance = Nitrogen intake − Nitrogen loss



- **Nitrogen balance equilibrium** is a state of the body in which the intake = output of nitrogen. This is found in the adults where growth has ceased and no more protein can be stored.



- **Positive nitrogen balance:** is a state of the body in which the intake > the loss of nitrogen from the body. This is the **body's anabolic state**.



- **Negative nitrogen balance:** is a state of the body in which the amount of nitrogen excreted > nitrogen ingested. This is the **body's catabolic state.**



Factors influencing nitrogen balance:

1. Growth:

- During the period of active growth → **Positive N.B**. exists.

2. Hormones:

- Growth hormones, insulin and androgens → **Positive N.B.,** while corticosteroids → **Negative N.B**.
- Thyroid hormone: hypothyroidism → **Positive N.B.**, hyperthyroidism → **Negative N.B.**

3. Injury, disease, and stress:

- Surgery, infection, burns, trauma, sepsis, and cancer → **Negative N.B.**

4. Psychological stress:

- Severe/prolonged stress → **Negative N.B.**

5. Physical exercise:

- Ordinary exercise → **Little or no effect**
- Intensive training/athletes → **Positive N.B.** *(particularly with adequate protein and energy intake)*

6. Diet and nutritional status:

- Malnutrition → **Negative N.B.**
- Adequate intake during nutritional rehabilitation → **Positive N.B.**

7. Pregnancy and lactation:

- **Positive N.B.**

The Recommended Dietary Allowance (RDA):

- **For healthy adults with normal body weight**, protein requirements are calculated using healthy body weight: **0.8 g/kg/day** **or**
- **The AMDR for protein10-35%** this intake should maintain body in state of nitrogen balance.
- An additional **25 g/day** is needed for pregnancy and lactation.
- Protein requirements of **infants and children** vary according to

Once protein needs are met; excess protein is used for energy or converted to fat and stored

Consumption of high protein diet (especially animal proteins) associated with :

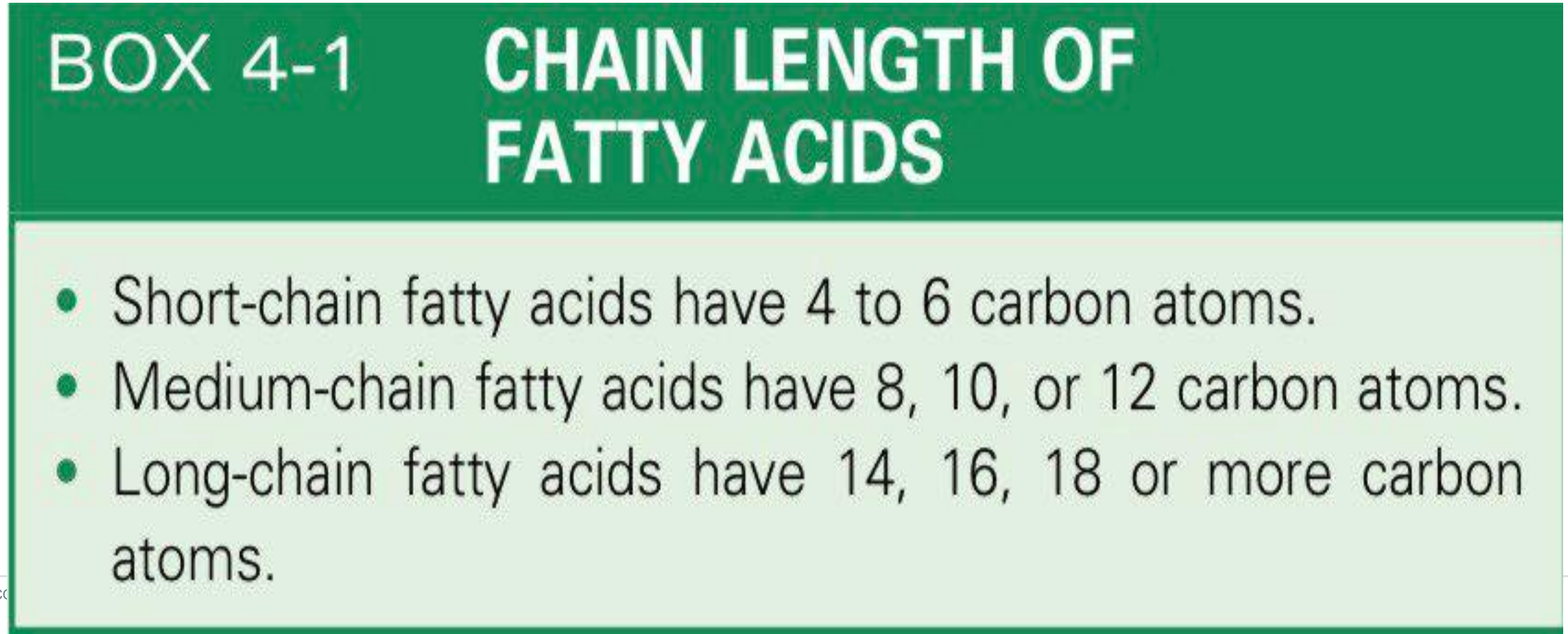
- **Heart Disease:** Protein itself is not known to increase the risk of heart disease or mortality. However, some protein-rich foods, particularly high-fat and processed meats such as lunch meats and hot dogs, may increase the risk of obesity, heart disease, and diabetes

- **Decline in kidney function:** High protein intake may accelerate the decline in kidney function, particularly in people with pre-existing mild kidney impairment.
- **Increased urinary calcium excretion:** Excessive protein intake, may increase urinary calcium excretion, which may contribute to bone loss in some circumstances.
- **Increased risk of kidney stone formation:** High protein intake may increase urinary calcium and uric acid excretion, potentially increasing the risk of kidney stone formation.

Lipids

Lipids divided according to their chemical properties of fatty acids:

1-Length of f. a. chain: the longer the chain more solid the fat.



2-Degree of saturation:

a-Saturated :

- **SFA** usually found in **animal foods** e.g.; fat in meat, butter , milk, eggs…. Among plant-based foods, **coconut oil and palm oil** are relatively high in saturated fatty acids.
- Solid at room temperature **(fat).**

b-Unsaturated f. a.:

- Generally, from **plant sources.**
- liquid at room temperature **(oil).**

I- monounsaturated (MUFA) : contain only **one double** bound

Oleic acid =omega 9 (olive oil, olive, avocado...).

II- Polyunsaturated (PUFA): contain **more than one double** bond.

1. Omega 3 FA (alpha linolenic acid, ALA). Found mainly in **flaxseed oil, chia seeds**, and some other plant foods. **Fatty fish** are rich mainly in the omega-3 fatty acids **EPA (Eicosapentaenoic acid)** and **DHA (Docosahexaenoic acid).**

2. Omega-6 FA (linoleic acid, LA): Found mainly in **vegetable cooking oils**, such as **corn oil and safflower oil**.

Types of EFAs:

Type of fatty acids	Type of chain	examples
w-3 Fatty Acids	Short chain	a-Linolenic acid
	Long chain	Eichosapentanoic acid Docosahexaenoic acid
w-6 Fatty Acids	Short chain	Linoleic Acid
	Long chain	Arachidonic acid gama-Linolenic acid

Omega 3 and omega 6 are called essential fatty acid(EFA)

Importance of Essential Fatty Acids:

1. Integrity of **cell membrane** structure and function especially in the retina and CNS.
2. **Regulate blood pressure.**
3. **Lowering serum cholesterol level**
4. Essential for normal **growth and development.**
5. Maintaining of **skin integrity**
6. Regulation of **inflammation** and **immune function**.

Clinical feature of EFA deficiency: deficiency is **rare** but can occur **in infant** fed on EFA-deficient formula, patients receiving fat free parenteral feeding or those with malabsorption

1. **Dermatitis, generalize,scaly, dry skin & eczema**
2. **Alopecia**
3. **Thrombocytopenia**
4. **Failure to thrive** and in children
5. **Poor wound healing**
6. **Increased susceptibility to infections**.



A, Essential fatty acid deficiency. Patients receiving fat-free parenteral have developed biochemical abnormalities and skin lesions as shown B, Resolution in same patient after 2 weeks of treatment. (Dr. M.C. Riella, from Dr. *A Colour atlas and text of diet-related disorders*, ed 2, London, 1992, Mosby Book Europe Limited. By permission of Mosby International Ltd.)

- Polyunsaturated and monounsaturated fat have **the same** numbers of calories as saturated fat.
- **Fresh water fish & sea foods** contain predominant unsaturated fatty acid mainly omega 3 (the Dietary Guidelines for Americans recommend eating **8 to 12 ounces of seafood per week**).

Type of Fat	Main Source	State at Room Temperature	Effect on Cholestero Levels
MUFA	Olives; olive oil ،canola oil ، peanut oil; cashews, almonds, peanuts ،and most other nuts ؛ avocados	Liquid	Lowers LDL؛ raise HDL
PUFA)	Corn, soybean, safflower, and cottonseed oils	Liquid	Lowers LDL but with no effect on HDL (
SFA	Whole milk ،butter, cheese, and ice cream; red meat; chocolate ؛ coconuts, coconut milk, and coconut oil	Solid	Raises both LDL and HDL
Trans	Most margarines ؛vegetable shortening ؛partially hydrogenated vegetable oil ؛ deep-fried chips ؛many fast foods; most commercial baked goods	Solid or semi-solid	Raises LDL; lowers HDL

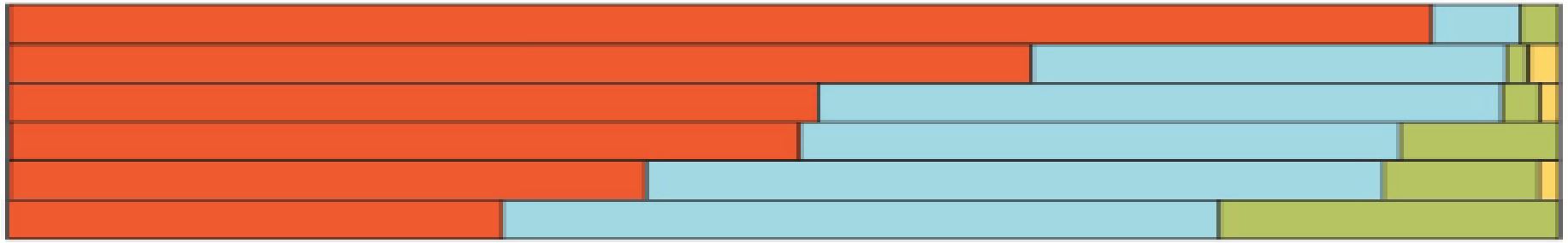
Most fats are a mixture of saturated, monounsaturated, and polyunsaturated fatty acids.

Key:

- Saturated fatty acids
- Monounsaturated fatty acids
- Polyunsaturated, omega-6 fatty acids
- Polyunsaturated, omega-3 fatty acids

Animal fats and the tropical oils of coconut and palm contain mostly saturated fatty acids.

Coconut oil
Butter
Beef tallow (beef fat)
Palm oil
Lard (pork fat)
Chicken fat



Some vegetable oils, such as olive and canola, are rich in monounsaturated fatty acids.

Olive oil
Canola oil
Peanut oil



Many vegetable oils are rich in omega-6 polyunsaturated fatty acids.

Safflower oil[a]
Sunflower oil
Corn oil
Soybean oil
Walnut oil
Cottonseed oil



Only a few oils provide significant omega-3 polyunsaturated fatty acids.

Flaxseed oil
Fish oil (salmon)



- **Cholesterol** is present only in foods of **animal origin** especially organs (liver, kidney, brain…) and **absent in plant foods.**
- **Cholesterol** is essential for **brain growth and development in children.**
- **Food sources** of cholesterol;

✓ Liver 400mg/3 oz serving.
✓ Lamb& beef 80-85mg/3oz serving.
✓ Egg 252mg (1 large).
✓ Poultry 74mg/3 oz serving (skinless).
✓ Fish45-60mg/3 oz =
✓ Whole milk 34mg/cup
✓ Ice cream 54-98mg/cup
✓ Cheese 28mg/oz serving.

Functions of Fats in the Body:

1. Fats are the body's chief form of **stored energy**.
2. **Protein sparing** action for tissue synthesis.
3. **Transport & absorption** of fat-soluble vitamins & Ca.
4. **Fat pads** protect the body's internal organs from shock.
5. Fat beneath the skin **insulates** the body from extremes of temperature.
6. Fats form the major material of **cell membranes.**
7. Fats are converted to other **compounds**, such as hormones, bile, and vitamin D

How Much Fat the body Needs?

- No more than **30% of total calories daily** intake should come from fat, with **AMDR (20-35%)**
- **Less than 10%** from **saturated fat**
- **5-10%** from **polyunsaturated fat**
- **The remaining** should be come from **monounsaturated fat.**
- Recommended dietary cholesterol **less than 300 mg/day.**
- Keep trans fat intakes **as low as possible**

Summery:

- ✓ Glycogen is a storage form of energy for human, and animals. Starch foods come from plants. Fibers are found in all plant derived foods(soluble and insoluble)
- ✓ DRI recommendation of fiber 14 grams per 1000 kcalorie.
- ✓ Prebiotic: A non-digestible dietary carbohydrate that is selectively utilized by beneficial microorganisms in the colon. Probiotics: are live microorganisms, when administered in adequate amounts, confer a health benefit on the host.
- ✓ Glycemic Index: is a scale that ranks foods by how much they raise blood glucose levels.
- ✓ Essential, nonessential protein, protein complementarity complete & incomplete protein.
- ✓ Nitrogen balance: The difference between the amount of nitrogen intake and the amount of nitrogen excreted or lost from the body is called nitrogen balance
- ✓ The AMDR for protein10-35% this intake should maintain body in state of nitrogen balance.
- ✓ Saturated & unsaturated FA (mono, poly),poly → omega 3&omega 6 → EFA.
- ✓ Total calories daily intake should come from fat, with AMDR (20-35%)

References

1. Nutrition and Diet Therapy 10th edition
2. Williams' Basic Nutrition and Diet Therapy 15 edition
3. Williams' Essentials of Nutrition and Diet Therapy 11th edition



QUESTIONS & DISCUSSION





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